

Spam Email Detection System

Artificial Intelligence Laboratory Project Report

Submitted By: Fahim Abrar

ID: 0112320144

Course: Artificial Intelligence Laboratory(CSE 3812)

Section: A

Group: C

1. Abstract

The volume of unsolicited bulk emails, commonly referred to as "spam," continues to pose significant challenges to digital communication. Spam not only clutters inboxes but often serves as a delivery mechanism for malware and phishing attacks. The objective of this project is to develop and evaluate a machine learning-based text classification model capable of automatically distinguishing between legitimate communications ("ham") and spam.

By leveraging Natural Language Processing (NLP) techniques for feature extraction and comparing the performance of probabilistic and linear classifiers, this project aims to implement a highly accurate and scalable filtering solution. The final pipeline utilizes Term Frequency-Inverse Document Frequency (TF-IDF) vectorization coupled with Logistic Regression, yielding a classification accuracy of 98.84%.

2. Introduction & Problem Statement

In the modern digital ecosystem, electronic mail remains one of the primary modes of professional and personal communication. However, the open nature of email protocols makes them highly susceptible to abuse. Spam filtering is a binary classification problem where an algorithm must learn the underlying semantic and syntactic patterns that differentiate malicious or promotional text from genuine messages.

The core challenge lies in the unstructured nature of text data. Algorithms require numerical inputs to perform mathematical optimizations. Therefore, translating human language into a mathematically quantifiable format without losing contextual meaning is the first hurdle. Additionally, the chosen algorithm must be robust enough to handle high-dimensional feature spaces, as the English vocabulary used across thousands of emails is vast.

3. Dataset Description

The project utilizes a pre-labeled dataset provided in the file `spam_ham_dataset.csv`. The dataset encapsulates historical email data, representing diverse writing styles and subjects.

Dataset Specifications:

- **Total Samples:** 5,171 emails.
- **Features:** Raw text content (the body and subject of the email).
- **Target Variable (label_num):** A binary indicator where 0 signifies 'Ham' (Legitimate) and 1 signifies 'Spam'.

The data was imported and managed using the `pandas` library, allowing for efficient tabular manipulation and verification of the underlying class distributions.

4. Methodology and Data Preprocessing

4.1 Text Vectorization (TF-IDF)

To transform the raw textual data into a machine-readable numerical format, the Term Frequency-Inverse Document Frequency (TF-IDF) algorithm was applied using the `TfidfVectorizer` from the `scikit-learn` library.

Unlike basic term-frequency approaches that merely count word occurrences, TF-IDF calculates a continuous statistical weight for each word. It rewards words that appear frequently within a specific email (Term Frequency) but penalizes words that appear commonly across the entire dataset (Inverse Document Frequency), such as "and", "the", or "is". Furthermore, standard English stop-words were explicitly filtered out during this phase to enhance the signal-to-noise ratio.

Vectorization Output: Applying TF-IDF to the 5,171 emails resulted in a massive high-dimensional matrix consisting of **50,140 unique features** (vocabulary words).

4.2 Data Splitting

To ensure an unbiased evaluation of the model's predictive capabilities, the dataset was partitioned into two mutually exclusive sets using an 80/20 split:

- **Training Set:** 4,136 samples (80% of the data). This set was exclusively used to train the classifiers, allowing them to learn the statistical boundaries between ham and spam.
- **Testing Set:** 1,035 samples (20% of the data). This set was withheld during the training phase and utilized solely to measure the model's accuracy on unseen data.

A static random state (`random_state=42`) was utilized to guarantee the reproducibility of the splits across multiple executions.

4.3 Algorithm Selection

Two distinct classification algorithms were implemented to determine the optimal approach for this specific textual feature space:

1. **Multinomial Naive Bayes:** A probabilistic classifier heavily utilized in NLP tasks. It applies Bayes' theorem under the "naive" assumption of conditional independence between every pair of features given the value of the class variable.
2. **Logistic Regression:** A linear model used for binary classification. It calculates the probability of a binary outcome by fitting data to a logit function, optimizing a decision boundary across the high-dimensional feature space.

5. Results and Evaluation

After training both models on the 4,136 training samples, their performance was evaluated against the 1,035 withheld testing samples using the `accuracy_score` metric.

Model	Testing Accuracy
Multinomial Naive Bayes	92.17%
Logistic Regression (max_iter=1000)	98.84%

5.1 Comparative Analysis

While both models demonstrated strong predictive capabilities, **Logistic Regression substantially outperformed Multinomial Naive Bayes**, achieving near-perfect accuracy (98.84%).

This performance discrepancy can be attributed to the mathematical nature of the selected vectorization technique. Multinomial Naive Bayes is fundamentally designed to operate on integer feature counts (e.g., Bag-of-Words). While it functions with fractional TF-IDF weights, it violates its core discrete assumptions. In contrast, Logistic Regression excels at drawing linear decision boundaries through continuous, high-dimensional spaces. The continuous weightings generated by TF-IDF perfectly complemented the mathematical optimization process of the Logistic Regression model.

6. Application & Sample Predictions

To validate the real-world applicability of the optimized Logistic Regression model, a custom set of unseen, unlabelled emails was processed through the inference pipeline. The model successfully tokenized, vectorized, and classified the following edge-case scenarios:

```
--- Sample Predictions ---
```

Email: 'Lose 20 pounds in 10 days with our miracle pill! 100% guaranteed results or your money back. Order now before stock runs out!!!'

Prediction: Spam

Email: 'Your recent order for 'Radiohead - Kid A (Vinyl)' has been shipped and will arrive by Thursday. Track your package here.'

Prediction: Ham (Not Spam)

Email: 'Hi team, just a reminder that our project meeting is scheduled for tomorrow at 10 AM in the lab.'

Prediction: Ham (Not Spam)

Email: 'CONGRATULATIONS! You are the lucky winner of a free iPhone. Reply with your address to claim.'

Prediction: Spam

7. Conclusion

The objective of building an automated, highly accurate spam detection system for the Artificial Intelligence Lab was successfully realized. By systematically preparing textual data through TF-IDF vectorization and rigorously evaluating multiple machine learning algorithms, the project yielded a highly robust classification pipeline.

The empirical results demonstrated that for continuous, high-dimensional TF-IDF feature spaces, Logistic Regression (98.84%) provides superior decision boundaries compared to probabilistic models like Naive Bayes (92.17%). The model successfully isolated malicious and promotional terminology—such as "winner", "miracle", and "guaranteed"—from natural conversational language.

In conclusion, this project highlights the immense capability of supervised machine learning and natural language processing in solving pressing cybersecurity and data-filtration challenges. Future iterations of this project could further enhance robustness by exploring ensemble methods or deep learning architectures, such as Recurrent Neural Networks (RNNs) or Transformers, to capture sequential context beyond individual word frequencies.